

$$a) R_i = 1M // 63M = \frac{63M \times 1M}{64M} = 0,98M = 984,37k\Omega$$

$$R_{i2} = 10k // h_{ie} // 50k = 10k // 12k // 50k = \frac{50 \times 1,66}{51,66} = 1,60k\Omega$$

$$R_o = 4,7k$$

$$b) A_{v1} = \frac{v_{o1}}{v_i} = -g_m (4,7k // R_{i2})$$

$$= -2,16 \times 10^{-3} \left(\frac{4,7k // 1,60k}{1,19} \right)$$

$$A_{v1} = -2,57$$

$$A_{v2} = \frac{v_{o2}}{v_{o1}} = -\beta \left(\frac{\frac{2,16k}{4,7k // 4k}}{2k} \right) = -120 (1,08) = -129,6$$

$$v_s = \frac{v_i}{v_s} = \frac{R_i}{R_i + 20k} = \frac{984,37k}{984,37k + 20k} = 0,98$$

$$A_{vs} = \frac{v_o}{v_s} = \frac{v_i}{v_s} \cdot \frac{v_{o1}}{v_i} \cdot \frac{v_o}{v_{o1}} = \frac{v_i}{v_s} \cdot A_{v2} \cdot A_{v1} = 0,98 \cdot (-3,57) \cdot (-129,6)$$

$$A_{vs} = 326,41$$

Hocam sekti baska kısıda gırdım. Soruda belirtilmediğiniz için yekün yapım.

V_{01} çıkışını opamp teslemlenen opamp



$$V_{01} = 1 + \frac{2k}{2k} \cdot \left(\frac{2k}{2k+10k} \cdot V_i \right) = \frac{1}{3} V_i //$$

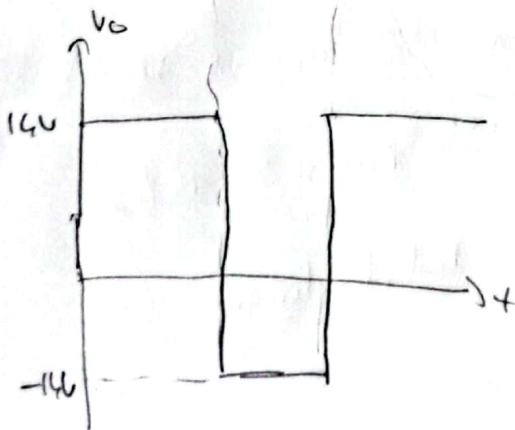
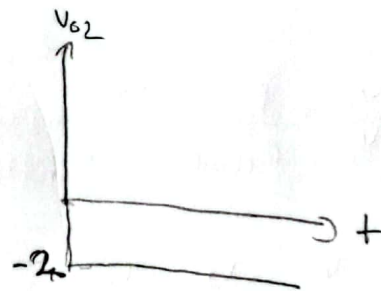
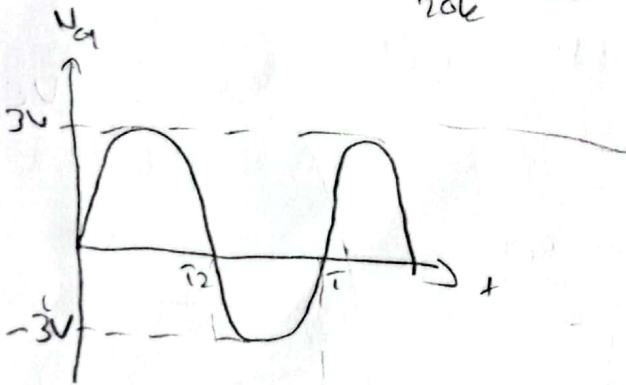
V_{02} çıkışlı opamp teslemlenen opamp.

$$V_{02} = \left(-\frac{4k}{4k} \cdot V_2 \right) + \left(-\frac{4k}{2k} V_3 \right)$$

$$V_{02} = 2 - 6 = -4V //$$

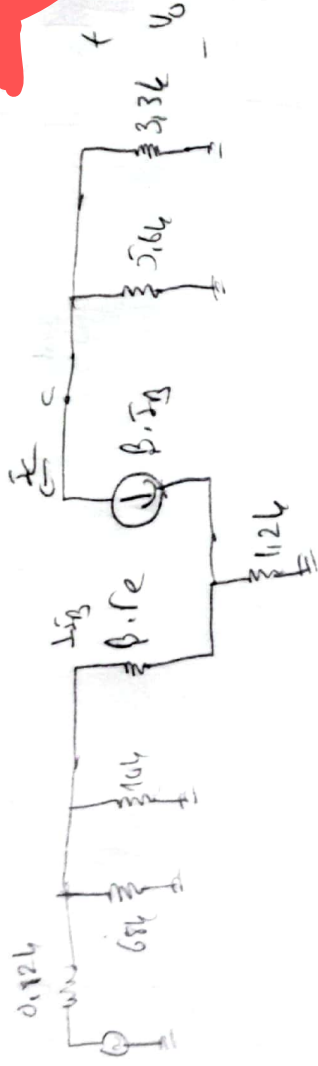
V_0 çıkışlı opamp teslemlenen opamp. Pozitif giriş $V_+ = \frac{1}{3} V_i$

Negatif giriş $\rightarrow V_- = \frac{10k}{20k} \cdot V_{02} = \frac{1}{2} V_{02} = \frac{1}{2} \cdot (-4) = -2V$



$$V_0 = \begin{cases} V_{cc} & : V_{01} \geq -2V \\ -V_{cc} & : V_{01} < -2V \end{cases}$$

3



$$5) \quad r_{i_b} = \frac{v_b}{i_b} = \frac{r_B \cdot \beta \cdot r_e + (\beta + 1)(r_B)(1.2k)}{\beta \cdot i_b}$$

$$r_{i_b} = \beta \cdot r_e + (\beta + 1)(1.2k) = (120 + 1)(1.2k) = 148.8k$$

$$r_i = (10k // 68k // r_{i_b}) = \frac{10k // 68k // 148.8k}{8.91} = 8.23k$$

$$r_o = 5.6k //$$

$$6) \quad A_v = \frac{v_o}{v_i} \rightarrow v_o = -\beta \cdot i_b (5.6k // 3.3k)$$

$$v_i = v_o = i_b \cdot r_{i_b}$$

$$\frac{-\beta (5.6k // 3.3k)}{\beta r_e + (\beta + 1)(1.2k)} = \frac{(5.6k // 3.3k)}{30 + 1.2k} = -\frac{2036}{1230} = -1.68$$

$$\frac{v_i}{v_s} = \frac{r_i}{r_i + 0.82k} = 0.907$$

$$A_{us} = A_v \cdot \frac{v_i}{v_s} = -1.58$$

3) d) 0,4 MF kapasitenin frekansı (siriştki)

$$f_1 = \frac{1}{2\pi \times 0,47 \times 10^{-6} (0,82k + 0,1)} = 37,4 \text{ kHz}$$



Giriştki $\rightarrow \frac{1}{2\pi \times 0,47 \times 10^{-6} (5,6k + 3,3k)} = 38,05 \text{ kHz}$

Emittördeki 20 MF kapasitesinin frekansı (kutup)

$$f_3 = \frac{1}{2\pi \cdot 20 \times 10^{-6} \left(1,2k \parallel \left(1,2k + \frac{\beta \cdot r_e + 10k \parallel 68k + 0,82k}{\beta + 1} \right) \right)}$$
$$= 108,5 \text{ Hz}$$

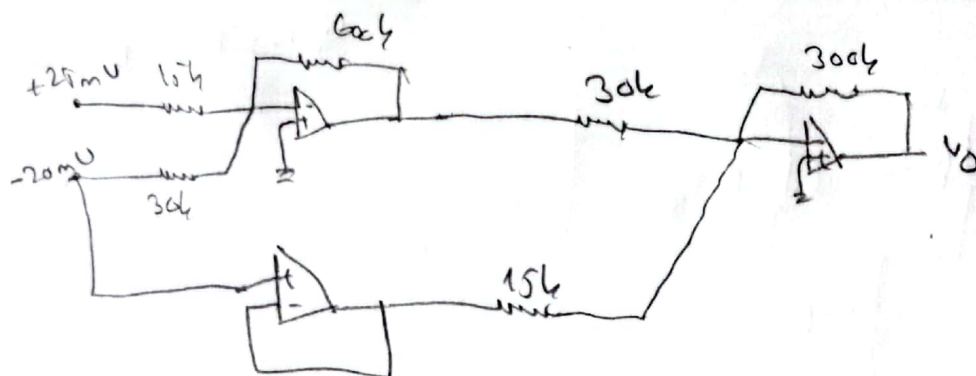
$$f_3 = \frac{1}{2\pi \times 20 \times 10^{-6} \times 625,1} = 12,73 \text{ kHz}$$

AH kesim frekansı (f_1 ve f_2 4ten aldyından birini okuyun)

$$f_c = \frac{f_2}{\sqrt{2^k - 1}} = \frac{38,05}{\sqrt{2^4 - 1}} = 59,45 \text{ kHz olur.}$$

0,64

4



V_{01} çıkış olan opamp eksen toplar.

$$V_{01} = \left(-\frac{60k}{15k} (25mV) \right) + \left(-\frac{60k}{30k} \cdot (-20mV) \right)$$

$$V_{01} = -1V + 0.4V = -0.6V$$

V_{02} çıkış opamp buffer dır.

$$V_{02} = -20mV \text{ dir}$$

V_0 çıkış opamp eksen toplar

$$V_0 = \left(-\frac{30k}{30k} \cdot V_{01} \right) + \left(-\frac{30k}{15k} \cdot V_{02} \right)$$

$$V_0 = (-10 \cdot (-0.6)) + (-20 \cdot (-20mV))$$

$$V_0 = 6 + 0.4 = 6.4V //$$